PubMed/PMC Task 1: Known-Item & Provenance Retrieval

Participants are tasked with developing an end-to-end system that accepts a complex biomedical information need, expressed as a natural-language question or instruction, and leverages a Challenge Corpus, comprising PubMed records and PubMed Central (PMC) full-text documents, to identify and retrieve the specific record(s)/document(s) that best satisfies the question. Up to five (5) candidate items will be accepted and considered as valid responses to questions for this task.

The following provides an example question and the corresponding expected response for this Challenge Task:

- **Question:** "What was the landmark paper that demonstrated the mechanistic discovery and potential to exploit CRISPR-Cas9 for RNA-programmable genome editing in vitro?"
- **Expected Response:** PMID: 22745249

Challenge Dataset

This challenge comes with its own Challenge Corpus that can be used in responding to the questions in this task. It consists of two constituent corpora:

PubMed

- The PubMed baseline files for the challenge are available at https://ftp.ncbi.nlm.nih.gov/pubmed/.pubmed-spark/
- This corpus includes all citations with a publication date from January 1,1959 to July 31,2026 and excludes any citation with the publication type "preprint" (MeSH UI D000076942)
- For more information about this dataset, see the relevant PubMed documentation page (https://pubmed.ncbi.nlm.nih.gov/download/).

PubMed Central (PMC)

- A subset of PMC shall be used as a source of full-text documents for responding to questions in this task. The resulting subset includes PMC publications that satisfy the specified Creative Commons licensing, NIH funding or intramural status, publication-date, and publication-status criteria.
- A list of PMCIDs defining the PMC component of the Challenge Corpus is available in the file named **"Challenge_PMCIDs.txt"**
- The corresponding PMC articles may be downloaded or accessed programmatically using NCBI/PMC-supported services, including the PMC Cloud Service, PMC OAI-PMH API, BioC API, NCBI E-Utilities, and PMC ID Converter API. The PMC Cloud Service supports bulk access to machine-readable article

files, while the OAI-PMH and BioC APIs support programmatic retrieval of article content. See https://pmc.ncbi.nlm.nih.gov/tools/developers/ Participants are responsible for complying with applicable article-level license and usage requirements. PMC identifies these services as the supported mechanisms for automated retrieval of PMC content.

- **PMC Cloud Service / AWS — Bulk Download**
  - The PMC Cloud Service provides a convenient mechanism for downloading the Challenge Corpus in bulk.
  - Available article files include JATS XML, plain text, metadata, and, where applicable, PDFs and supplementary materials.
  - The PMC datasets are accessible through AWS without requiring an AWS account or login.
  - Participants should use the provided PMCID list to identify and retrieve the articles comprising the PMC component of the Challenge Corpus.
  - For more information on the PMC Cloud Service and available datasets, please see: https://pmc.ncbi.nlm.nih.gov/tools/cloud/
  - For general information about PubMed Central, please see: https://pmc.ncbi.nlm.nih.gov/

Submission Structure

To be considered for evaluation in this challenge, participants must upload a submission file containing responses for up to 1,000 natural-language questions that make up the evaluation set for this task. On January 15 2027, Participants must submit a Docker image(s) of their solution, a description of their system, including its components, technical documentation, and deployment and execution instructions, in accordance with the System Submission requirements provided below.

Participants will be provided the evaluation set of questions on January 18, 2027. Subsequently, participants should develop responses to these needs and submit them via the challenge portal prior to or on January 31, 2027. Up to five (5) PubMed or PubMed Central items will be accepted and considered as candidate valid responses to each question for this task if all are appropriate; submissions will be penalized if they include incorrect citations. Participants can submit up to three (3) submissions for this task.

Submissions fille should be UTF-8-encoded and use JSONL file structure, with each line being a JSON object corresponding to an question and a response.

Each Submission Record consists of the following components:

- The metadata object identifies the participating team, system run, and question identifier and original question.
- The response object contains the system's retrieval results. The records array contains one or more retrieved publications, with each record identified by record_ID (either a PubMed ID, and or PubMed Central ID).

Here is a breakdown of the Submission Record for each question:

```
├── metadata
│   ├── team_id
│   ├── submission_id
│   ├── info_need_id
│   └── info_need
│
└── response # list of up to five PM/PMC IDs
```

Along with a short explanation here:

- **metadata** — Identifies the submission and question.
  - **team_id** — Unique identifier for the participating team.
  - **submission_id** — Identifies a specific system run or submission.
  - **info_need_id** — Unique identifier for the question.
  - **info_need**— The natural-language biomedical question.
- **response** — Array containing one or more (up to five) identifier(s) of the retrieved publication(s) (e.g., PMID or PMCID).

Sample Questions and Expected Task Responses

See PM-PMC-Task1-Sample-Submission-Simple.jsonl.txt

System Submission

By January 15, 2027, participants must submit a Docker image(s) containing the solution, a description of their system and its components, together with sufficient technical documentation and instructions to enable NIH to reproduce, deploy, and evaluate the system within an NIH-managed AWS environment. All software, models, indexes, data resources, documentation, and other materials that comprise or support the submitted system must have been created or last modified on or before January 15, 2027. A link to the participant's GitHub/Docker repository for the solution will be accepted.

Submissions should describe the system architecture and components, including retrieval, ranking, embeddings, indexes, models, LLM inference, and any other software or data resources required to execute the system. Participants may design their systems as a single containerized application or as multiple containerized services, including distributed architectures in which different components run on separate compute nodes. Participants should also provide deployment and execution instructions, along with an estimate of the computational and storage resources required (e.g., CPU, memory, GPU, and storage).

Participants will not be required to submit executable container images, source code, or large data artifacts (e.g., embeddings, indexes, or models) as part of this initial submission. These materials will be requested during the evaluation phase in accordance with the Challenge procedures and technical guidance.

For evaluation, NIH will provision and manage the underlying AWS infrastructure, including compute, storage, networking, and orchestration. Participants will not be required to provision or manage AWS resources, Kubernetes clusters, or Terraform infrastructure.

To support verification of the January 15, 2027 submission deadline, participants may be required to provide cryptographic hashes or other verifiable records for software, models, indexes, data artifacts, and other system components identified in their submission. When requested during the evaluation phase, the corresponding executable or data artifacts must match the versions documented and verified at the submission deadline.

Model and Resource Restrictions: Submitted systems may use only publicly available, open-weight Large Language Models (LLMs), open-source software, and/or other publicly available computing resources. Systems may not invoke external APIs to proprietary frontier models (e.g., commercial hosted LLM endpoints) or rely on non-public, restricted-access software, models, datasets, or other resources during execution. Any pretrained/fine-tuned models used by the system must be included within the submitted Docker images.

- Docker Image Execution and Network Restrictions
  - The challenge organizers reserve the right to run the submitted Docker-based system and verify system reproducibility, resource utilization, and compliance with challenge rules. Specifically:
  - External Calls: Submitted Docker Images will be executed in a completely isolated environment with all external network calls blocked. Limited access will be permitted to NLM-hosted services, such as the UMLS Terminology Services (UTS), and other

challenge specific services. Details of these approved services and endpoints will be made available when the challenge is published. This zero-egress policy (excluding the permitted services) prevents both data leaks and runtime reliance on external computing resources, tools, or services.

Additional technical guidance, including submission formats, deployment specifications, resource limits, verification requirements, and evaluation procedures, will be provided to participants sufficiently in advance of the January 15, 2027 submission deadline.

Evaluation and Challenge Winner Selection

Phase 1: System Selection

For Phase 1 Evaluation, a sample of about 200 questions (from the 1,000 questions) will be used to identify the top-performing systems for a more extensive evaluation in Phase 2. Participant responses, consisting of up to five (5) ranked items for each question, will be compared against expert-verified reference responses. Note, only one these ranked items can be (potentially) correct.

The primary evaluation metric for this task will be Mean Reciprocal Rank (MRR) across the questions. MRR measures how highly the system ranks the one relevant item for each question and rewards systems that place the relevant item at the top of their ranked results.

For each question, the Reciprocal Rank (RR) is calculated as the reciprocal of the rank position of the highest-ranked relevant item:

RR = 1 / rank of the relevant item

For example, if the relevant item is ranked first, the RR is 1.0; if it is ranked second, the RR is 0.5; if it is ranked third, the RR is 0.33; and so forth. If none of the retrieved items is relevant, the RR is 0.

The MRR is the mean of the Reciprocal Rank scores across all questions:

- MRR = average RR across all questions

The top N (expected to be five) systems based on Phase 1 MRR performance will advance to Phase 2.

Phase 2: Extended Evaluation and System Deployment

The selected systems will be evaluated on a private set of questions designed to assess retrieval across a broad range of biomedical questions, including complex requests requiring identification of specific publications and claims involving historical or scientific priority.

In this phase, each of the semifinalist systems will be installed and configured in an NIH-provided AWS cloud environment to ensure a standardized evaluation environment. These systems will be evaluated using the private set of questions, ease of system installation, resource usage (memory, disk, compute), and response time.

Phase 2 will use the following evaluation criteria:

- **MRR:** Measures how highly a system ranks the relevant item for each question, rewarding systems that place the relevant item near the top of their ranked results.
- **Ease of Use:** Assesses the effort required to deploy, configure, operate, and use the system in the standardized NIH cloud environment.
- **Compute Time:** Measures the computational time required to generate responses.
- **Compute Effort:** Measures the computational effort required to generate responses, including the resources and processing required by the system.
- **Memory Footprint:** Measures the memory and other computational resources required to operate the system and generate responses.

MRR will serve as the primary metric for determining final system rankings and identifying the overall winning systems. In the event of a tie in MRR, the additional evaluation criteria listed above may be used to differentiate systems and determine the final ranking.